

2017 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Collisions, momentum, impulse & uncertainty

Two 0.180 kg snooker balls collide head-on: white +2.60 m s⁻¹, black -1.80 m s⁻¹; afterwards black is +2.38 m s⁻¹.

(a)(i) Final velocity of the white ball

Conservation of momentum (equal masses so divide by m): $u_{\text{white}} + u_{\text{black}} = v_{\text{white}} + v_{\text{black}}$.

$$v_{\text{white}} = 2.60 + (-1.80) - 2.38 = -1.58 \text{ m s}^{-1} \text{ (left).}$$

(a)(ii) Inelastic collision

Momentum is conserved but kinetic energy is not; some KE is transformed to heat, sound, deformation.

(b)(i) Average force by the cue

Impulse: $F_{\text{avg}} = m\Delta v/\Delta t = (0.180 \times 0.84)/0.040 = 3.78 \text{ N}$.

(b)(ii) Percentage uncertainty in F

For $F = m v / t$, percentage uncertainties add: $\Delta F/F \approx \Delta m/m + \Delta v/v + \Delta t/t$.

$$= (0.001/0.180) + (0.01/0.84) + (0.001/0.040) = 0.0425 \Rightarrow 4.3\%.$$

Revision tips

- Track directions with signs in momentum equations.
- Inelastic: KE not conserved, momentum is.
- Impulse is area under F-t curve and equals Δp .
- When multiplying/dividing measured quantities, add percentage uncertainties.